Continuing on writing the method section of the report. is this section factually correct?

The constructed bars contain information about the traded price and volume (how many shares are traded), but that itself is not sufficient for generating an edge in algorithmic trading. That's why we introduce \textit{feature engineering}. These features are solely calculated from the OHLCV data we gather from Yahoo Finance. The model derives 36 different features, amongst which are Relative Strength Index, Bollinger Bands, Chaikin Money Flow, and more. These features are given to the model in addition to the bars during training, testing, and validating

Mostly, but not fully. The indicator examples are fine, but a few details are either too strong or not supported by the attached sources.[1] [2]

Issues

- Saying raw bars are “not sufficient for generating an edge” is too absolute; the safer claim is that raw OHLCV bars may be **insufficient on their own** or may not capture enough market structure for your model.[1]
- If this paragraph is based on the attached crypto paper, “Yahoo Finance” is not correct, because that paper uses Binance tick-level data and then computes features after the sampling procedure.[1]
- “Solely calculated from the OHLCV data” is not fully correct if your feature set includes sine/cosine encodings of hour and weekday, because those are time-based features rather than OHLCV-derived indicators.[1]
- “36 different features” is not verified by the attached sources; the crypto paper refers to 33 input variables in one model setup, while the ResNet paper uses close-price sequences rather than a 36-feature OHLCV feature set.[2] [1]
- “Traded price and volume” is imprecise for bars; a bar normally contains open, high, low, close, and volume, and the crypto paper explicitly says the sampled point is still a typical OHLC bar with a non-fixed period.[1]

Better wording

A safer version would be: “The constructed bars contain open, high, low, close, and volume information. To enrich this representation, we apply feature engineering and compute additional indicators such as RSI, Bollinger Bands, and Chaikin Money Flow. These features are computed after bar construction and provided to the model during training, validation, and testing.”[1]

If your implementation truly uses Yahoo Finance and exactly 36 engineered features, then keep those two details, but I would still change “not sufficient” to “may not be sufficient” and “validating” to “validation.”[1]

⁂

we aim to use a range of different bars, cusum, dollar etc

Then phrase it around **information-driven sampling**, not just “constructed bars.”[3]

You could write: “Rather than relying only on fixed time bars, we consider several information-driven sampling methods, including dollar bars, volume bars, range bars, and the CUSUM filter, which generate new bars when market activity or price movements reach predefined thresholds.”[3]

If you want to keep the feature-engineering part, a clean version is: “Each sampled observation is still represented as an OHLCV bar, but its duration is not fixed in clock time. We then compute additional features from these OHLCV bars and provide them to the model during training, validation, and testing.”[3]

Small terminology fix: CUSUM is usually called a **filter** rather than a bar type, so “sampling methods including dollar bars, volume bars, range bars, and the CUSUM filter” is the most accurate wording.[3]

⁂

these dollar, volume and cusum bars are part of the informtion driven sampling?

Yes. In the attached paper, **information-driven sampling** explicitly includes volume bars, dollar bars, range bars, and the CUSUM filter.[4]

The small terminology nuance is that dollar bars, volume bars, and range bars are bar types, while CUSUM is more accurately a **filter** that triggers when a threshold is exceeded.[4]

So a clean sentence is: “We use information-driven sampling methods, including dollar bars, volume bars, range bars, and the CUSUM filter.”[4]

⁂

We believe that this method of \textit{information driven sampling} will help us achieve better than market returns, because it's fundamentally different from techniques used in current literature. Algorithmic trading literature predominantly focuses on data sampling based on regular time intervals \cite{gradzki2025crypto}. This seems fundamentally unaligned with how financial markets operate today, where there events that effect the stock price flow continously.

Is my part about why we use information driven sampling factually correct?

Mostly yes, but two parts are too strong. The attached paper supports the claim that algorithmic trading literature predominantly relies on regular time-interval sampling and that information-driven sampling is motivated by market activity and volatility rather than fixed clock time.[5]

What is correct

- Saying the literature mainly uses regular time intervals is supported; the paper states that algorithmic trading literature relies predominantly on aggregation based on regular time intervals, and its review notes that a vast majority of studies use systematic time-based sampling such as hourly or daily bars.[5]
- Saying this can be misaligned with market behavior is also supported; the paper argues that trades occur continuously, that round-hour intervals have no inherent significance, and that information-driven methods adapt in real time to market dynamics.[5]

What to change

- “We believe that this method … will help us achieve better than market returns” is not a safe factual statement for a methods section; it is better framed as a motivation or hypothesis, not as an established fact.[5]
- “Because it's fundamentally different from techniques used in current literature” is a bit overstated; safer wording is that it differs from the dominant or conventional approach in the literature, not all current literature.[5]

Better version

You could write: “We use information-driven sampling because the algorithmic trading literature predominantly relies on fixed time-interval aggregation, even though financial markets evolve continuously and trading activity is uneven over time. Information-driven methods such as dollar bars, volume bars, range bars, and the CUSUM filter instead sample observations when market activity or price movement reaches predefined thresholds, which may provide a representation of market dynamics that is more aligned with real trading conditions."[5]

Also fix the sentence to "where there are events that affect price formation continuously," or even better, "where trading activity and price changes occur continuously."[5]

⁂

find some littereature about tripple barrier method

Yes — here are a few good references to cite for the **triple barrier method.**[6] [7] [8]

Core sources

- **Marcos López de Prado,** *Advances in Financial Machine Learning* **(2018)** is the standard foundational reference most later explainers point to for the triple barrier method and its three barriers: profit-taking, stop-loss, and a vertical time limit.[7] [8] [6]
- **Grądzki, Wójcik, and Lessmann (2025), "Algorithmic crypto trading using information-driven bars, triple barrier labeling and deep learning"** is a strong applied paper because it directly compares Triple Barrier labeling with next-bar prediction in an algorithmic trading setting.[9]

Practical literature

- **Interactive Brokers / QuantInsti (2024), "The Triple Barrier Method: A Python GPU-based Computation"** is useful for explaining the intuition of the method in plain language, especially the role of volatility-adaptive profit-taking and stop-loss levels plus the vertical barrier.[6] [7]
- **William Santos (2022), "Algorithmic trading: triple barrier labelling"** is helpful if you want an implementation-oriented source, since it explains the labeling logic and discusses computational complexity.[10]

Applied papers

- **"Stock Price Prediction Using Triple Barrier Labeling and Raw OHLCV Data" (arXiv)** is relevant if you want a modern deep-learning application of Triple Barrier labeling on stock data rather than just a conceptual description.[11]
- The attached **crypto paper** is especially useful for your report because it argues that Triple Barrier labeling is more practical than next-bar prediction and better reflects trading with stop-loss and take-profit orders.[9]

A compact sentence you can use is: "We adopt the Triple Barrier method, originally popularized by López de Prado, because it defines labels through profit-taking, stop-loss, and time-based exit rules, making it more aligned with real trading decisions than simple next-bar labeling."[8] [7] [6] [9]

⁂

Explain whats written in Marcos Lopez de Prado's work

In López de Prado's *Advances in Financial Machine Learning* as described by the sources here, the **Triple Barrier Method** is a labeling method for financial ML that frames each event like a trade with predefined exit rules rather than simply asking whether the next bar goes up or down.[15] [16]

Core idea

The method places three barriers around an entry point: an upper barrier for profit-taking, a lower barrier for stop-loss, and a vertical barrier that limits how long the trade can stay open.[16] [15]

The label is then determined by which barrier is hit first, so the outcome reflects both price movement and time.[15] [16]

Why he uses it

The motivation is to make labels look more like real trading decisions, because traders usually manage positions with upside targets, downside limits, and a maximum holding period.[16] [15]
This is why later applied work describes Triple Barrier labeling as more practical than next-bar prediction and more aligned with stop-loss and take-profit trading logic.[15]

Important details

The sources also describe the method as adaptive rather than fixed, since the profit-taking and stop-loss levels are typically scaled to market volatility instead of using arbitrary constant thresholds.[16] [15]
They further note that related utilities such as daily volatility estimation and label filtering are presented as being based on López de Prado's book.[16]

A simple one-sentence summary for your report is: "López de Prado's Triple Barrier method labels each event according to whether a volatility-scaled profit-taking barrier, stop-loss barrier, or time barrier is reached first, making the target closer to an actual trading outcome than standard next-bar labeling."[15] [16]

⁂

@misc{santos2022triplebarrier,

title = {Algorithmic trading: triple barrier labelling},
year = {2022},
url = {https://williamsantos.me/posts/2022/triple-barrier-labelling-algorithm/},
note = {Accessed: 2026-03-11}
}

make a .bib for the source

Use this BibTeX entry:[17]

```
@misc{santos2022triplebarrier,
  author      = {William Santos},
  title       = {Algorithmic trading: triple barrier labelling},
  year        = {2022},
  url         = {https://williamsantos.me/posts/2022/triple-barrier-labelling-algorithm
  note        = {Accessed: 2026-03-11}
}
```

⁂

so each bar essentially gets a trading decision label?

Yes, **roughly** each event or bar gets a label, but more precisely each bar is treated as a potential trade entry and is labeled according to which barrier is hit first afterward.[26]

So the label is not "what this bar is," but rather "what would have happened if a position had been opened at this bar."[26]

A clean way to say it in your report is: "Under the Triple Barrier method, each bar is treated as a candidate entry point and assigned a trading-outcome label based on whether the profit-taking barrier, stop-loss barrier, or time barrier is reached first."[26]

⁂

\subsection{Triple Barrier Method}

The Triple Barrier Method is the next step in the data pipeline. This is also a part of the preprocessing of data that we perform before training starts. It's a method that places three barriers for an entry point on the stock, and it's an attempt to make the bars look like potential trading decisions. It is called a triple barrier because each bar has one of three optimal trading decisions associated with it: Sell for upper barrier profit-taking, sell for lower-barrier stop-loss, and sell due to a limit set for how long one can hold the stock \cite{santos2022triplebarrier}.

Am i correct in writing this?

Almost, but not fully. The main idea is right, but "optimal trading decisions" and the repeated use of "sell" are not accurate descriptions of the Triple Barrier Method.[27] [28] [29]

Fixes

The method is better described as a **labeling** procedure that assigns a label to each candidate entry point according to which of three barriers is hit first: profit-taking, stop-loss, or a vertical time barrier.[28] [27]
So it is not that each bar has three "optimal trading decisions"; rather, each bar is treated as a hypothetical trade entry and then labeled by its outcome.[30] [28]

Also, the upper barrier is not simply "sell," and the lower barrier is not always "sell" either; more generally, they represent a profit-taking outcome and a stop-loss outcome, while the third outcome is time expiry if neither price barrier is reached first.[29] [27] [28]

Better wording

You could write:

```
\subsection{Triple Barrier Method}
The Triple Barrier Method is the next step in the data pipeline and forms part of the da
```

This version is factually closer to how the method is described in the literature.[27] [28]

⁂

do they assume you have bought the stock when assessing the triplle barriers?

Not necessarily. The Triple Barrier Method does not inherently assume you have already bought the stock; more generally, it labels what happens from a candidate **entry** under a profit-taking barrier, a stop-loss barrier, and a maximum holding period.[38] [39]

If you apply it in a long-only setup, then yes, it effectively means "assume we enter long here" and then check whether profit-taking, stop-loss, or the time barrier is hit first.[40] [39]

So for your report, the safest wording is: "Each bar is treated as a potential entry point, and the label reflects the outcome of that hypothetical position."[39] [38]

⁂

elaborate on the next bar method

The **next-bar method** is a simpler labeling approach where each observation is assigned a target based on what happens in the immediately following bar, usually whether the next bar's price moves up or down relative to the current one.[46]
In other words, the model is trained to predict the direction of the next sampled interval, not the outcome of a full trade with exit rules.[46]

How it works

If bar t is the current bar, the label is determined from bar $t + 1$, for example by checking whether the next closing price is above or below the current price.[46]
This makes the method easy to implement, which is why it is widely used in the literature.[46]

Limitation

Its main weakness is that it fixes the prediction horizon to exactly one bar, even though real trades are usually managed with stop-losses, take-profits, and varying holding times.[46]
The attached paper argues that this can make next-bar prediction less realistic for trading and harder to turn into profit after transaction costs, especially at short horizons.[46]

Compared with Triple Barrier

Next-bar labeling asks, "What will the next bar do?"[46]
Triple Barrier labeling instead asks, "If I entered here, would profit-taking, stop-loss, or time expiry happen first?"[46]

A clean sentence for your report is: "The next-bar method labels each observation according to the direction of the immediately following bar, making it a simple and common prediction target, but one that may not fully reflect real trading outcomes."[46]

⁂

you cant do these bar labelling method during inference right since you dont have access to the following bar information

Yes — not in live inference the same way you do during dataset construction. Both next-bar labeling and Triple Barrier labeling require **future** information, such as the following bar or the path of prices until a profit-taking, stop-loss, or time barrier is reached.[56] [57] [58]

So these labeling methods are mainly for creating targets offline during preprocessing and training, not for assigning a true label at the current live timestamp.[58] [56]

At inference time, the model only sees information available up to "now" and predicts which label is most likely; the **actual** label can only be known later, once future price movement has occurred.[57] [56]

A clean way to write it is: "Triple Barrier and next-bar labels are ex-post targets used for supervised learning, since their true values depend on future price evolution and therefore are unavailable at inference time."[56] [58]

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